11 Chemistry

Unit 2 Revision Checklist

Use this checklist to plan your revision for the end-of-semester examination. For each statement, tick how confident you feel. Then spend most of your time on the 'Not yet' and 'Nearly' rows: use the revision worksheet, your notes and the practice examination to work on them, and tick again later.

Tick one box per row: Got it · Nearly · Not yet

The examination: 10 minutes reading time and 90 minutes writing time. Section A is multiple choice (answered on a separate answer sheet); Section B is short answer. You will have the VCE Chemistry Data Book and a scientific calculator.

1. Solution concentration

Study design: solution concentration as a measure of the quantity of solute dissolved in a given mass or volume of solution (mol L^{-1}, g L^{-1}, % (m/v)), including unit conversions

I can…	Got it	Nearly	Not yet
define concentration as the amount of solute in a given volume of solution			
calculate a concentration in mol L^{-1} from a mass of solute and a volume of solution ($n = m \div M$, then $c = n \div V$)			
calculate a concentration in g L^{-1} and in % (m/v)			
convert between mol L^{-1}, g L^{-1}, mg L^{-1} and % (m/v) in either direction			
convert volumes between mL and L before using $c = n \div V$			
find the concentration of one ion in a solution of an ionic compound (e.g. using the ratio in the formula)			

2. Standard solutions and dilution

Study design: calculations related to the preparation of standard solutions and the dilution of solutions

I can…	Got it	Nearly	Not yet
describe what a standard solution is and the properties of a suitable primary standard			
calculate the mass of solute needed to make a standard solution of a given volume and concentration			
describe how to prepare a standard solution			
use $c_1V_1 = c_2V_2$ to find a new concentration, the volume of stock needed, or the volume of water to add			
work through two or more dilution steps by multiplying the dilution factors			
choose suitable glassware for preparing and diluting solutions			

3. Volumetric analysis (titrations)

Study design: volume–volume stoichiometry (solutions only) and the use of acid–base titrations (no back titrations) to find the concentration of an acid or a base

I can…	Got it	Nearly	Not yet
name the equipment used in a titration and describe how each piece is prepared before use			
use the terms aliquot, titre, end point and concordant titres correctly			
calculate titres from initial and final burette readings			
identify concordant titres and calculate an average titre			
write a balanced equation for a neutralisation reaction and use its mole ratio			
calculate an unknown concentration from titration data, including when the ratio is not 1:1			
predict whether a procedural error makes the calculated result too high, too low or unchanged, and explain why			

4. Greenhouse gases

Study design: CO_2, CH_4 and H_2O as three of the major gases that contribute to the natural and enhanced greenhouse effects due to their ability to absorb infrared radiation

I can…	Got it	Nearly	Not yet
name the three major greenhouse gases in the study design			
explain how greenhouse gases warm the atmosphere			
explain the difference between the natural and the enhanced greenhouse effect			
identify human activities that add each greenhouse gas to the atmosphere			
interpret data on greenhouse gas concentrations			

5. Gas pressure and SLC

Study design: the definitions of gas pressure and standard laboratory conditions (SLC) at 25 °C and 100 kPa

I can…	Got it	Nearly	Not yet
explain changes in gas pressure using the behaviour of gas particles			
state the conditions of SLC and find the molar volume at SLC in the Data Book			
convert pressures between kPa, Pa and atm (1 atm = 101.3 kPa)			

6. The ideal gas equation

Study design: calculations using pV = nRT, limited to kPa, Pa, atm, mL, L, °C and K (including unit conversions)

I can…	Got it	Nearly	Not yet
use pV = nRT with p in kPa, V in L and T in K (so R = 8.31 J mol^{-1} K^{-1})			
convert °C to K (add 273) and mL to L before substituting			
rearrange pV = nRT to find any one of p, V, n or T			
convert a calculated temperature back to °C when the question asks for it			
predict how pressure, volume or temperature changes when another variable changes			
know when to use 24.8 L mol^{-1} (only at SLC) and when to use pV = nRT instead			

7. Gas stoichiometry

Study design: the use of stoichiometry to solve calculations related to chemical reactions involving gases (including moles, mass and volume of gases)

I can…	Got it	Nearly	Not yet
convert between mass, amount (mol) and gas volume for any gas in a reaction			
use the mole ratio from a balanced equation to go from one substance to another			
calculate the volume of a gas produced (or used) at SLC or at other conditions			
work through a multi-step calculation that combines a gas volume with a mass or a solution concentration			

8. Molar volume and molar mass of a gas

Study design: calculations of the molar volume or molar mass of a gas produced by a chemical reaction

I can…	Got it	Nearly	Not yet
calculate molar volume from a measured gas volume and the amount of gas ($V_m = V \div n$)			
calculate the molar mass of a gas from its mass and volume at known conditions ($M = m \div n$)			
use a molar mass to identify an unknown gas			
explain how an experimental error would affect a calculated molar volume or molar mass			

9. Limiting and excess reactants

Study design: limiting and excess reactants

I can…	Got it	Nearly	Not yet
explain what limiting and excess reactants are			
identify the limiting reactant using the balanced equation			
calculate the amount or mass of product from the limiting reactant			
calculate the amount or mass of the excess reactant left over			
explain how changing the amount of one reactant affects the amount of product			

10. Specific heat capacity and calorimetry

Study design: specific heat capacity of water including units and symbols, and calorimetry calculations

I can…	Got it	Nearly	Not yet
use the specific heat capacity of water and its units correctly			
use q = mcΔT, with ΔT as a temperature change (not a single temperature)			
rearrange q = mcΔT to find m, ΔT or a final temperature			
convert between J and kJ, and give energy released or absorbed as a positive value			
calculate the energy released or absorbed per gram or per mole of a substance			
identify the limiting reactant before calculating energy per mole in a reaction			
identify the assumptions made in a calorimetry experiment and evaluate how they affect a result			

11. Redox reactions and equations

Study design: oxidising and reducing agents, and redox reactions, including writing balanced half and overall redox equations (including in acidic conditions), with states

I can…	Got it	Nearly	Not yet
define oxidation and reduction as loss and gain of electrons			
assign oxidation numbers and use them to identify the species oxidised and reduced			
identify the oxidising agent and the reducing agent in a reaction			
write simple half-equations for metals, metal ions and non-metals, including states			
balance a half-equation in acidic conditions using water, hydrogen ions and electrons			
combine half-equations into a balanced overall equation with no electrons left over			

12. Reactivity series and displacement

Study design: the reactivity series of metals and metal displacement reactions, including balanced redox equations with states

I can…	Got it	Nearly	Not yet
use the electrochemical series in the Data Book to predict whether a displacement reaction will occur			
describe observations for a displacement reaction (deposits forming, colour changes)			
write half and overall equations for displacement reactions, including states			
use experimental results to rank metals as reducing agents			
design a fair test to compare the reactivity of metals			

13. Redox in society: corrosion and primary cells

Study design: applications of redox reactions in society: for example, corrosion or the use of simple primary cells to produce electrical energy from chemical energy

I can…	Got it	Nearly	Not yet
describe the conditions needed for iron to corrode, and the redox reactions involved			
describe ways of preventing corrosion			
describe a simple primary (galvanic) cell and identify where oxidation and reduction occur			
describe the movement of electrons and ions, and the energy transformation, in a primary cell			
use the electrochemical series or experimental data to predict which metal pairs give a larger voltage			

14. Exam skills

I can…	Got it	Nearly	Not yet
show all working, set out one step per line			
give final answers with units and an appropriate number of significant figures (match the data; round only at the end)			
include states in every equation			
use the VCE Chemistry Data Book for molar masses, SLC, R, the specific heat capacity of water and the electrochemical series			
match my answer to the command word: ‘state/identify’ needs one point; ‘explain’ needs a reason and a link back to the question			
pace myself at about 1.15 minutes per mark			

My three priorities for revision:

1. _____

2. _____

3. _____